AXONBRIDGE

Universal Agentic Automation Layer

by Axonic Health

Document Suite
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DOCUMENT 1

Concept Note

AxonBridge: The AI Agentic Automation Layer for Healthcare

The Problem

Hospitals worldwide have invested heavily in Hospital Information Systems (HIS), Electronic Health Records (EHR), and legacy administrative platforms. These systems are deeply embedded — switching costs are catastrophic, change management is politically and operationally complex, and vendor contracts are long-term. Yet these same systems are slow, siloed, and unable to deliver the AI-augmented workflows that clinical and administrative staff urgently need.

The result: hospitals are locked in. They cannot access the next generation of AI capability without replacing the very systems that hold their institutional memory. This is not a problem unique to healthcare — it affects insurance, pharmaceuticals, government health portals, and any regulated industry built on legacy software.

The AxonBridge Solution

AxonBridge is a Universal Agentic Automation Layer — an AI-powered middleware that operates on top of any existing application, exactly as a skilled human operator would, without requiring API access, vendor cooperation, or system replacement.

Powered by browser automation agents (vision-based AI, screen understanding, and natural language processing), AxonBridge perceives any application interface, understands clinical and administrative context, and executes intelligent actions — all through the existing UI. The underlying HIS does not change. The hospital changes nothing. But every workflow becomes AI-augmented.

Without AxonBridge	With AxonBridge
Manual data entry across systems	Agents auto-populate fields across any HIS
Clinicians spending 40% of time on documentation	AxonScribe captures, understands, and files automatically
AI tools require expensive HIS integration projects	AI sits on top — no integration project needed
Each HIS upgrade breaks existing integrations	Agent layer adapts dynamically to UI changes
Multilingual access requires bespoke localisation	140+ language support embedded at the agent layer

Multilingual by Design

A core architectural principle of AxonBridge is language-first design. Healthcare's most underserved populations are those who face both technology and language barriers simultaneously. AxonBridge integrates multilingual NLP and voice processing at every layer of the stack:

- Voice input in the clinician's or patient's native language is transcribed, understood, and actioned in real time
- On-screen content from any HIS is translated and re-presented to the operating agent in the working language

- Outputs — documentation, summaries, alerts — are generated in the required output language
- Support for 140+ languages including low-resource Indic, Arabic, Malay, Swahili, and European languages
- Dialect and code-switching awareness for mixed-language clinical environments (e.g. Hindi-English, Hinglish)

Strategic Positioning

AxonBridge is not a replacement for existing HIS or EHR systems. It is the intelligence layer that works with whatever system the customer already has. This removes the single biggest barrier to AI adoption in healthcare: the change management problem.

The go-to-market entry point is healthcare, where Axonic Health has proven clinical credibility, 70+ hospital deployments, and deep domain knowledge across India, UK, and GCC markets. The architecture, however, is domain-agnostic — the same automation infrastructure applies to insurance, pharma, and government health portals in subsequent phases.

Immediate Commercial Opportunity

- Hospitals with existing HIS contracts seeking AI capability without system replacement
- Multi-facility healthcare groups running fragmented, inconsistent HIS estates post-merger
- Insurance and payor organisations with high-volume manual pre-authorisation and claims workflows
- Government health programmes requiring AI workflow automation across legacy portals in multiple languages
- Medical education institutions requiring AI simulation and documentation on existing platforms

Axonic Health's Unique Unfair Advantage

No RPA or browser automation company possesses what Axonic brings to this market:

- A practicing NHS Consultant Surgeon as CEO — clinical credibility at the point of sale
- 70+ hospital deployments and 10M+ patient transactions — real deployment experience
- AxonMD's 15+ specialty AI bots — the intelligence layer is already built
- AxonScribe and AxonHIS — deep understanding of clinical documentation and HIS data models
- BAPIO network of 12,000+ UK doctors — distribution and trust channel
- Existing multilingual infrastructure across 140 languages — a capability competitors must build from scratch

DOCUMENT 2

Recommended Technology Stack

The AxonBridge stack is designed around four principles: reliability in clinical environments, language-first architecture, human-in-the-loop safety for clinical actions, and domain intelligence over generic automation. Each layer is selected to minimise vendor lock-in while maximising capability.

Layer 1 — Perception & Automation Engine

This is the foundation: the capability to see, understand, and interact with any application UI without API access.

Browser & Desktop Automation	
Playwright (Microsoft)	Primary browser automation framework. Handles headless/headed Chrome, Firefox, WebKit. Production-grade, actively maintained, handles dynamic SPAs. Open source.
Puppeteer (Google)	Secondary/fallback for Chromium-specific tasks. Widely used, large community support.
PyAutoGUI / pywinauto	Desktop GUI automation for thick-client legacy HIS applications (Windows-based). Handles non-browser applications.
Skyvern (open source)	Purpose-built browser agent using vision + LLM for dynamic web UI interaction. Handles forms that break traditional selectors. Strong candidate for core agent loop.

Vision & Screen Understanding	
GPT-4o Vision / Claude 3.5 Vision	Multimodal LLMs for interpreting screenshots, understanding form context, identifying UI elements that cannot be reliably selected by DOM traversal.
OpenCV + Tesseract OCR	Lightweight screen region detection and text extraction for structured forms. Faster and cheaper than LLM calls for routine field extraction.
YOLO v8 (Ultralytics)	Real-time UI element detection model for high-speed screen parsing in performance-critical workflows.

Layer 2 — Multilingual NLP & Voice Engine

Language capability is not an add-on — it is embedded at every interaction point. The architecture handles input (voice/text), in-context translation (for agent understanding), and output generation in the target language.

Speech-to-Text (STT)

OpenAI Whisper (large-v3)	Primary STT. Supports 99 languages including Hindi, Arabic, Malay, Swahili. Handles accented English well. Can be self-hosted for data sovereignty in India/GCC.
Google Speech-to-Text v2	Fallback and supplementary for real-time streaming where Whisper latency is a constraint. Strong Indic language support.
Azure Cognitive Services Speech	For NHS/UK deployments where Microsoft ecosystem compliance is required.

Translation & Language Understanding	
IndicTrans2 (AI4Bharat)	State-of-the-art open-source translation model for 22 scheduled Indian languages. Essential for rural India deployments. Self-hostable.
Helsinki-NLP / OPUS-MT	Open-source translation for 140+ language pairs. Lightweight, hostable on-premise.
Google Cloud Translation API	Commercial fallback for language pairs not covered by open-source models. Used where real-time quality is critical.
LangDetect + fastText	Automatic language identification at input. Handles code-switching (Hinglish, Manglish) via character n-gram models.

Text-to-Speech (TTS)	
Coqui TTS / XTTS-v2	Open-source multilingual TTS with voice cloning capability. Self-hostable. Supports Indian, Arabic, and Southeast Asian languages.
Google Cloud TTS	Commercial fallback. Best-in-class naturalness for patient-facing interactions.
Azure Neural TTS	NHS-preferred for UK deployments. SSML support for clinical terminology pronunciation control.

Layer 3 — AI Reasoning & Clinical Intelligence

This is where AxonBridge differentiates from generic RPA tools. The agent does not merely automate — it understands clinical context, applies specialty-specific knowledge, and operates within established care protocols.

Core LLM Reasoning	
Claude 3.5 Sonnet (Anthropic)	Primary reasoning engine for clinical workflow planning, intent classification, and multi-step task decomposition. Preferred for safety-critical contexts.
GPT-4o (OpenAI)	Secondary LLM for vision-heavy tasks and where OpenAI function-calling patterns offer workflow advantages.
Llama 3 70B (Meta, self-hosted)	On-premise option for deployments where data cannot leave the hospital network (India government, GCC sovereign cloud requirements).

Agent Orchestration

LangGraph (LangChain)	Stateful agent workflow orchestration. Handles multi-step clinical tasks with branching logic, error recovery, and human-in-the-loop checkpoints.
AutoGen (Microsoft)	Multi-agent framework for tasks requiring cooperation between specialist agents (e.g., documentation agent + clinical decision agent + billing agent).
Temporal.io	Durable workflow execution engine for long-running automation tasks. Handles retries, timeouts, and audit trails — essential for clinical compliance.

Domain Knowledge — AxonMD Integration	
AxonMD Specialty Bots (15+)	Existing Axonic clinical AI bots integrated as tool-calling endpoints within the agent's reasoning loop. Provides specialty-specific clinical decision support.
AxonScribe Clinical NLP	Existing Axonic clinical documentation engine. Processes transcripts into structured clinical notes, SOAP format, ICD-10/SNOMED coding.
Clinical Knowledge Graphs	Structured medical knowledge (drug interactions, diagnostic pathways, clinical protocols) embedded as retrieval-augmented context for agent reasoning.

Layer 4 — Data, Security & Compliance Infrastructure

Data Storage & Management	
PostgreSQL + pgvector	Primary relational store with vector extension for embedding-based retrieval of patient context, workflow history, and knowledge base.
Redis	In-memory cache for agent state, session context, and real-time workflow coordination.
MinIO (S3-compatible)	On-premise object storage for clinical documents, screen recordings, and audit artefacts. Self-hostable for data sovereignty.
FHIR R4 Layer (HAPI FHIR)	Standards-compliant health data exchange layer. Future-proofs AxonBridge for API-first HIS migration; allows graceful upgrade from UI automation to native integration.

Security & Compliance	
OAuth 2.0 / SAML 2.0	Enterprise identity and access management. Integrates with hospital Active Directory and government identity frameworks (India: ABDM, UK: NHS Login).
HashiCorp Vault	Secrets management for API keys, credentials, and certificate rotation in multi-tenant deployments.
Audit Log Service (immutable)	Tamper-proof audit trail of every agent action — timestamp, screenshot, data read/written. Required for NABH, JCI, CQC, and GDPR compliance.
End-to-End Encryption (AES-256)	All data in transit and at rest. Clinical data never leaves the hospital network in on-premise deployments.

Layer 5 — Deployment & Operations

Deployment Infrastructure	
Docker + Kubernetes	Containerised deployment for cloud, hybrid, and on-premise environments. Enables rapid hospital onboarding without infrastructure dependency.
Helm Charts	Repeatable, version-controlled deployment configurations for each HIS target environment.
Kong API Gateway	API management, rate limiting, and routing across multi-tenant hospital deployments.
Prometheus + Grafana	Real-time monitoring of agent performance, error rates, task completion, and latency — with clinical workflow-specific dashboards.

Open Source First

Rationale: The core automation infrastructure (Playwright, Whisper, IndicTrans2, LangGraph, HAPI FHIR) is built on open-source components. This eliminates vendor lock-in at the infrastructure layer, reduces ongoing licensing cost, supports on-premise deployment for data-sovereign markets (India government, GCC), and allows Axonic to retain the commercial layer as its proprietary asset.

DOCUMENT 3

Functional Requirements Document (FRD)

AxonBridge — Universal Agentic Automation Layer

Document Reference: AXB-FRD-001
Version: 1.0
Status: Draft — For Review
Scope: Phase 1 — Healthcare (HIS/EHR Automation); Multilingual Support; Human-in-the-Loop Clinical Safety
Out of Scope: Non-healthcare verticals (Phase 2+); Full clinical action autonomy without human confirmation; Consumer-facing deployment

1. System Overview

AxonBridge is a multi-layer software platform that places an intelligent automation agent between a human operator (clinician, administrator, or data entry staff) and any existing application. The system perceives the application UI, understands the clinical or administrative context, executes structured workflows, and surfaces AI-generated recommendations — all without modifying or requiring cooperation from the underlying application.

The system operates in two primary modes:

- Assisted Mode: Agent observes and suggests; human confirms every action before execution. Default for all clinical workflows.
- Automated Mode: Agent executes predefined, low-risk administrative workflows autonomously with post-hoc audit. Subject to explicit hospital administrator configuration.

2. Functional Requirements

FR-01 — UI Perception & Application Interaction

Req ID	Priority	Requirement
FR-01.1	MUST	System must perceive and interact with any web-based application running in a standard browser without requiring source code access, API credentials, or vendor cooperation.
FR-01.2	MUST	System must perceive and interact with thick-client (Windows desktop) HIS applications using GUI automation where browser-based access is unavailable.
FR-01.3	MUST	System must identify form fields, buttons, dropdowns, tables, and navigation elements using a combination of DOM analysis and computer vision, with vision as fallback when DOM is obfuscated or dynamically rendered.
FR-01.4	MUST	System must handle dynamically rendered Single Page Applications (SPAs), AJAX-loaded content, and session-state-dependent UI without selector breakage.

FR-01.5	SHOULD	System should detect UI changes (application updates, layout changes) and alert the administrator rather than failing silently.
FR-01.6	SHOULD	System should capture and store screenshots of every agent interaction for audit purposes, with configurable retention period.
FR-01.7	COULD	System could support mobile application automation (Android/iOS) via Appium integration for mobile-first HIS deployments in low-resource settings.

FR-02 — Multilingual Support

Req ID	Priority	Requirement
FR-02.1	MUST	System must support voice input in a minimum of 30 languages at launch, expanding to 140+ languages within 12 months, including Hindi, Tamil, Telugu, Bengali, Marathi, Gujarati, Punjabi, Kannada, Malayalam, Arabic (Modern Standard and Gulf dialects), Bahasa Malay, Swahili, and UK English variants.
FR-02.2	MUST	System must auto-detect the input language without requiring explicit user selection.
FR-02.3	MUST	System must handle code-switching (mixed-language input, e.g., Hinglish, clinical English within a Hindi sentence) and correctly extract clinical entities regardless of which language they appear in.
FR-02.4	MUST	System must produce clinical documentation output in the language specified by the hospital administrator, regardless of input language.
FR-02.5	MUST	System must translate HIS screen content captured via vision into the agent's working language for internal reasoning, without storing the translation unless required for audit.
FR-02.6	MUST	System must support text-to-speech output for patient-facing interactions in the patient's detected or selected language.
FR-02.7	MUST	All system prompts, alerts, and operator-facing messages must be configurable in the local language of the deployment.
FR-02.8	SHOULD	System should support right-to-left (RTL) text rendering for Arabic, Urdu, and Hebrew in operator-facing interfaces.
FR-02.9	SHOULD	System should include dialect-aware processing for Arabic (distinguishing Gulf Arabic from MSA in GCC deployments) and regional Hindi variants.
FR-02.10	COULD	System could support sign language input via gesture recognition for accessibility compliance in UK NHS deployments.

FR-03 — Clinical Documentation Automation

Req ID	Priority	Requirement
FR-03.1	MUST	System must transcribe spoken clinical consultations in real time and structure the transcript into a clinical note (SOAP format: Subjective, Objective, Assessment, Plan).
FR-03.2	MUST	System must extract clinical entities from transcribed text: diagnoses, medications (including dosage and frequency), allergies, procedures, and vital signs.

FR-03.3	MUST	System must map extracted clinical entities to relevant coding standards: ICD-10, SNOMED CT, and LOINC. NHS deployments must also support Read codes.
FR-03.4	MUST	System must populate structured fields in the target HIS with extracted and coded data, presenting the populated form to the clinician for review before submission.
FR-03.5	MUST	System must flag any data item that exceeds a confidence threshold for mandatory human review before submission.
FR-03.6	MUST	System must never submit clinical data to the HIS without explicit clinician confirmation in Assisted Mode.
FR-03.7	SHOULD	System should generate structured discharge summaries, referral letters, and outpatient clinic letters from consultation transcripts, in the appropriate professional register for the target market.
FR-03.8	SHOULD	System should detect potential prescribing errors, drug-drug interactions, and allergy conflicts during medication entry, surfacing alerts before HIS submission.
FR-03.9	COULD	System could generate patient-facing summaries in plain language, in the patient's preferred language, suitable for sharing via AxonCare mobile app.

FR-04 — Administrative Workflow Automation

Req ID	Priority	Requirement
FR-04.1	MUST	System must automate appointment scheduling workflows across any HIS that exposes a scheduling interface, including slot availability queries, booking, and confirmation generation.
FR-04.2	MUST	System must automate patient registration workflows: capturing demographic data from voice or document input and populating all relevant HIS registration fields.
FR-04.3	MUST	System must automate billing and charge capture workflows: extracting billable items from clinical documentation and populating HIS billing modules.
FR-04.4	MUST	System must support cross-system data retrieval: querying data from one application (e.g., lab system) and presenting it in context within another application's workflow.
FR-04.5	SHOULD	System should automate insurance pre-authorisation workflows across common payor portals used in India and GCC, including document attachment and status tracking.
FR-04.6	SHOULD	System should automate bed management and ward allocation workflows for inpatient contexts.
FR-04.7	COULD	System could automate government regulatory reporting submissions (e.g., notifiable disease reporting, PCPNDT compliance in India).

FR-05 — Clinical Decision Support Integration

Req ID	Priority	Requirement

FR-05.1	MUST	System must invoke AxonMD specialty bots as tool-calling endpoints within the agent workflow, surfacing relevant clinical decision support at the point of documentation.
FR-05.2	MUST	System must display clinical decision support outputs (differentials, recommended investigations, treatment guidelines) as a non-obstructive overlay within the existing HIS workflow.
FR-05.3	MUST	System must not make autonomous clinical decisions. All clinical decision support is advisory; documentation of whether advice was accepted or overridden must be captured.
FR-05.4	SHOULD	System should surface contextually relevant clinical guidelines (NICE, WHO, ICMR) based on the active diagnosis or medication being recorded.
FR-05.5	SHOULD	System should flag patients meeting criteria for preventive care protocols (vaccination schedules, screening programmes) based on demographic and clinical data visible in the HIS.

FR-06 — Human-in-the-Loop & Safety Controls

Req ID	Priority	Requirement
FR-06.1	MUST	All clinical data submissions (medications, diagnoses, procedures) must require explicit human confirmation before execution. No autonomous clinical data entry.
FR-06.2	MUST	System must provide a one-click abort mechanism that halts all active agent actions immediately and restores the HIS to its pre-action state where technically possible.
FR-06.3	MUST	System must display a clear, real-time visual indicator when an agent action is in progress, distinguishing agent activity from human operator activity on screen.
FR-06.4	MUST	System must maintain a complete, immutable audit log: every action proposed by the agent, human decision (accept/modify/reject), and final data submitted to the HIS.
FR-06.5	MUST	Automated Mode (no human confirmation) must be restricted to a pre-approved list of administrative workflow types, configurable by the hospital administrator with a default-deny policy.
FR-06.6	SHOULD	System should implement a confidence scoring mechanism: actions below a configurable confidence threshold are always escalated to human review regardless of mode.
FR-06.7	SHOULD	System should detect anomalous patterns (e.g., unusual prescription volumes, out-of-range vital signs) and require supervisor confirmation before allowing these data points to be submitted.

FR-07 — Workflow Learning & Adaptation

Req ID	Priority	Requirement
FR-07.1	MUST	System must support a Workflow Recording mode in which an administrator performs a task manually and the system captures the steps as a replayable automation template.

FR-07.2	MUST	Recorded workflow templates must be editable by an administrator without requiring developer involvement.
FR-07.3	MUST	System must adapt to minor UI changes in the target application (button repositioning, field label changes) without requiring full workflow re-recording.
FR-07.4	SHOULD	System should learn from human corrections: when a clinician modifies an agent-suggested data entry, the correction should be used to improve future suggestions for that hospital's context.
FR-07.5	COULD	System could surface workflow efficiency analytics: identifying bottlenecks, most-common manual corrections, and automation success rates by workflow type.

FR-08 — Integration & Interoperability

Req ID	Priority	Requirement
FR-08.1	MUST	System must support deployment alongside any HIS without requiring any modification to the HIS itself.
FR-08.2	MUST	System must expose a FHIR R4-compliant API for hospitals that wish to receive structured data output from AxonBridge into their own data infrastructure.
FR-08.3	MUST	System must integrate with AxonCare patient app to deliver patient-facing outputs (appointment confirmations, discharge summaries, medication reminders) in the patient's language.
FR-08.4	SHOULD	System should support HL7 v2 message output for legacy HIS environments that accept HL7 feeds but do not expose modern APIs.
FR-08.5	SHOULD	System should integrate with India's ABDM (Ayushman Bharat Digital Mission) for ABHA-linked health record creation and retrieval.
FR-08.6	SHOULD	System should integrate with NHS systems including NHS Login, the NHS Spine, and EMIS/SystmOne for UK deployments.
FR-08.7	COULD	System could integrate with wearable and IoT device data streams (AxonHealthHub Health ATM) to pre-populate vital signs into HIS workflows.

FR-09 — Security, Privacy & Compliance

Req ID	Priority	Requirement
FR-09.1	MUST	All patient data processed by AxonBridge must be encrypted in transit (TLS 1.3) and at rest (AES-256).
FR-09.2	MUST	System must support on-premise deployment in which no patient data leaves the hospital network. Cloud processing must be configurable as optional only.
FR-09.3	MUST	System must comply with applicable data protection regulations: GDPR (UK/EU), DPDP Act 2023 (India), and applicable GCC health data regulations (UAE MOHAP, Saudi NCA).
FR-09.4	MUST	System must support role-based access control (RBAC) with granular permissions: clinician, administrator, auditor, and system roles.

FR-09.5	MUST	System must support multi-factor authentication and integrate with hospital SSO/Active Directory.
FR-09.6	MUST	System must not use patient data for model training without explicit, documented, informed consent and appropriate DPA.
FR-09.7	SHOULD	System should obtain NABH digital health certification compliance for India deployments and DSP Toolkit compliance for NHS deployments.

3. Non-Functional Requirements

Category	Target	Notes
Response Time (Assisted Mode)	< 2 seconds	Time from voice input completion to suggested action display
Response Time (Automated Mode)	< 500ms	Time from trigger event to action execution for pre-approved workflows
STT Accuracy (English)	> 95% WER	Word error rate on clinical vocabulary benchmark dataset
STT Accuracy (Indic Languages)	> 88% WER	Hindi, Tamil, Telugu on clinical vocabulary benchmark
System Availability	99.5% uptime	Measured monthly; excludes planned maintenance windows
Concurrent Sessions	500+ per instance	Horizontally scalable via Kubernetes; no hard ceiling at architecture level
Audit Log Retention	7 years	Aligned with NHS and NABH record retention standards; configurable per jurisdiction
UI Adaptation Time	< 4 hours	Time to recover from HIS minor UI update without full workflow re-recording
Onboarding Time (new HIS)	< 5 days	From initial access to first live workflow template for a standard HIS

4. System Architecture — Component Summary

Component	Module	Function
Perception	UIPerceiver	Screen capture, DOM analysis, vision fallback, element identification
Perception	AppConnector	Browser/desktop driver management, session handling, HIS target library
Language	VoiceEngine	STT, language detection, code-switch handling, TTS output
Language	TranslationRouter	Input→working language; output→target language; dialect management

Language	ClinicalNLP	Entity extraction, ICD/SNOMED coding, SOAP structuring (AxonScribe)
Intelligence	AgentOrchestrator	Multi-step task planning, tool routing, state management (LangGraph)
Intelligence	ClinicalReasoner	AxonMD bot invocation, guideline retrieval, clinical alert generation
Intelligence	WorkflowEngine	Template execution, recording mode, adaptation logic
Safety	HumanInTheLoop	Confirmation UI, confidence scoring, abort controller
Safety	AuditService	Immutable action log, screenshot archive, compliance reporting
Integration	FHIRGateway	Structured data output, ABDM integration, NHS Spine connector
Infrastructure	TenantManager	Multi-hospital isolation, RBAC, SSO, configuration management

5. Phase 1 Delivery Milestones

Phase	Timeline	Deliverable
M0–M2	Foundation	UIPerceiver + AppConnector operational on 2 target HIS (eHospital, Practo). Playwright + vision fallback. Basic workflow recording.
M2–M4	Language Core	Whisper STT for English + Hindi + Arabic. IndicTrans2 integration. Basic TTS. Language detection and code-switch handling.
M4–M6	Clinical Intelligence	AxonScribe integration for SOAP documentation. ICD-10 coding. AxonMD bot invocation. Human-in-the-loop confirmation UI.
M6–M9	Safety & Compliance	Full audit logging. FHIR R4 output. RBAC and SSO. On-premise deployment packaging. NABH/DSP Toolkit alignment.
M9–M12	Scale & Language Expansion	30+ language STT/TTS. 5+ HIS targets. Automated Mode for administrative workflows. Performance optimisation to NFR targets.
M12+	Platform Maturity	140+ languages. Self-adapting workflow templates. Analytics dashboard. Adjacent vertical pilots (insurance, pharma).

6. Key Risks & Mitigations

Risk	Severity	Mitigation
HIS UI changes break automation workflows	HIGH	Vision-based fallback reduces selector fragility. UI change detection alerts administrator within 1 hour. Workflow recovery target < 4 hours.
HIS vendor legal challenge to screen automation	MEDIUM	AxonBridge reads publicly rendered UI output — legally equivalent to a human operator. No reverse engineering, no

		memory access. Legal opinion obtained for key markets before commercial launch.
Clinical data error due to agent misclassification	HIGH	Assisted Mode default. Confidence scoring. All clinical data requires explicit human confirmation. Full audit trail. Errors traceable and correctable.
Low STT accuracy for rare Indic dialects	MEDIUM	IndicTrans2 supplemented by Google STT fallback. Clinical vocabulary fine-tuning per deployment. Continuous correction feedback loop.
Data sovereignty constraints in India/GCC government deployments	HIGH	On-premise deployment packaging. No patient data leaves hospital network. Self-hosted LLM option (Llama 3 70B). DPDP and GCC health regulation compliance documented.
Agent reliability in high-concurrency hospital environments	MEDIUM	Kubernetes horizontal scaling. Redis session isolation. Load testing against 500+ concurrent session target before production release.

Axonic Health — Building the Intelligence Layer for Global Healthcare

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